ISOM3360

Data Mining for Business Analytics

Ensemble methods

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Department of ISOM

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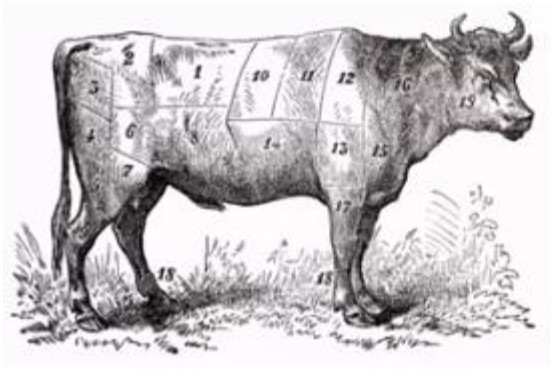


Wisdom of Crowds

- Human ensembles are demonstrably better
  - ❖Two heads are better than one.



- Classical example
  - ❖Estimate the weight of an ox (Francis Galton, 1906)





average of 800 guesses = 1,197
actual weight of the ox = 1,198

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Ensemble Learning in Practice: Netflix Prize



Leaderboard

Showing Test Score. Click here to show quiz score

Rank	Team Name	Best Test Score	% Improvement	Best Submit Time
Grand Prize - RMSE = 0.8567 - Winning Team: BellKor's Pragmatic Chaos				
1	BellKor's Pragmatic Chaos	0.8567	10.06	2009-07-26 18:18:28
2	The Ensemble	0.8567	10.06	2009-07-26 18:38:22
3	Grand Prize Team	0.8582	9.90	2009-07-10 21:24:40
4	Opera Solutions and Vandelay United	0.8588	9.84	2009-07-10 01:12:31
5	Vandelay Industries !	0.8591	9.81	2009-07-10 00:32:20
6	PragmaticTheory	0.8594	9.77	2009-06-24 12:06:56
7	BellKor in BigChaos	0.8601	9.70	2009-05-13 08:14:09
8	Dace_	0.8612	9.59	2009-07-24 17:18:43
9	Feeds2	0.8622	9.48	2009-07-12 13:11:51
10	BigChaos	0.8623	9.47	2009-04-07 12:33:59
11	Opera Solutions	0.8623	9.47	2009-07-24 00:34:07
12	BellKor	0.8624	9.46	2009-07-26 17:19:11

It was important to utilize a variety of models that complement the shortcomings of each other.

-- Lessons from Netflix Prize Challenge

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Ensemble Learning

- What is ensemble learning?
  - ❖A collection of methods that predict a target variable by training a number of models and combining their predictions.

- Why do we use ensemble learning?
  - ❖An ensemble of models is generally more accurate than any of its individual members.



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Constructing Ensembles to Achieve Diversity

- Diversity from different algorithms, or algorithm hyper-parameters
- Diversity from differences in inputs
  - ❖Divide up training data
  - ❖Reweight training data
  - ❖Different feature selections/weightings



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Types of Ensembles

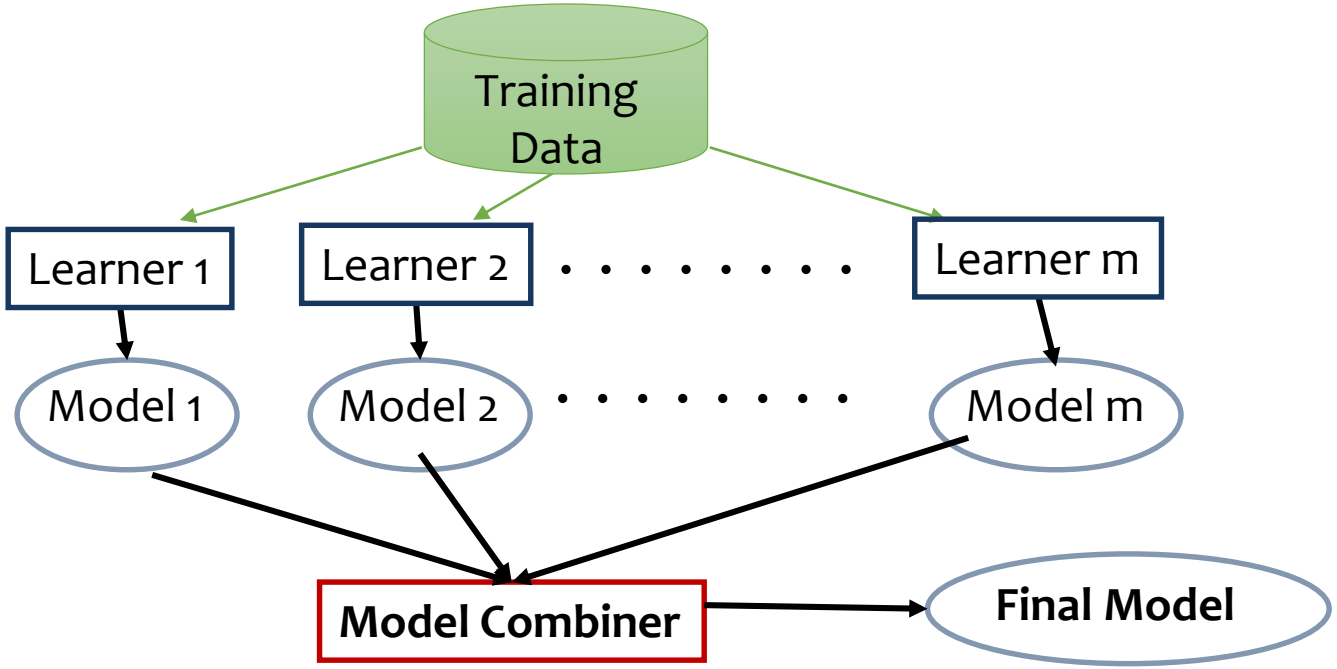
- Simple averaging
  - ❖Train multiple models by applying different learning algorithms on the same data, then combine their predictions
- Bagging
  - ❖Train multiple models by applying one classifier on different bootstrap samples, then combine their predictions
- Boosting (e.g., AdaBoost)
  - ❖Train multiple models by applying one classifier on different reweighted data, then combine their predictions



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Simple Averaging

- Learn multiple models using different learning algorithms.
  - ❖Different learners: e.g., 1 nearest neighbor model + 1 naïve Bayes model
  - ❖Same learner with different hyper-parameters: e.g., 5 neighbor-based models with different numbers of neighbors



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Simple Averaging Predictions

- In classification, we combine the predictions from different models by taking:
  - ❖The most popular prediction
  - ❖The proportion of predictions in each class
  - ❖Weighted voting

- In regression, we combine the predictions from different models by taking:
  - ❖The average
  - ❖The median
  - ❖Weighted average



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Bagging

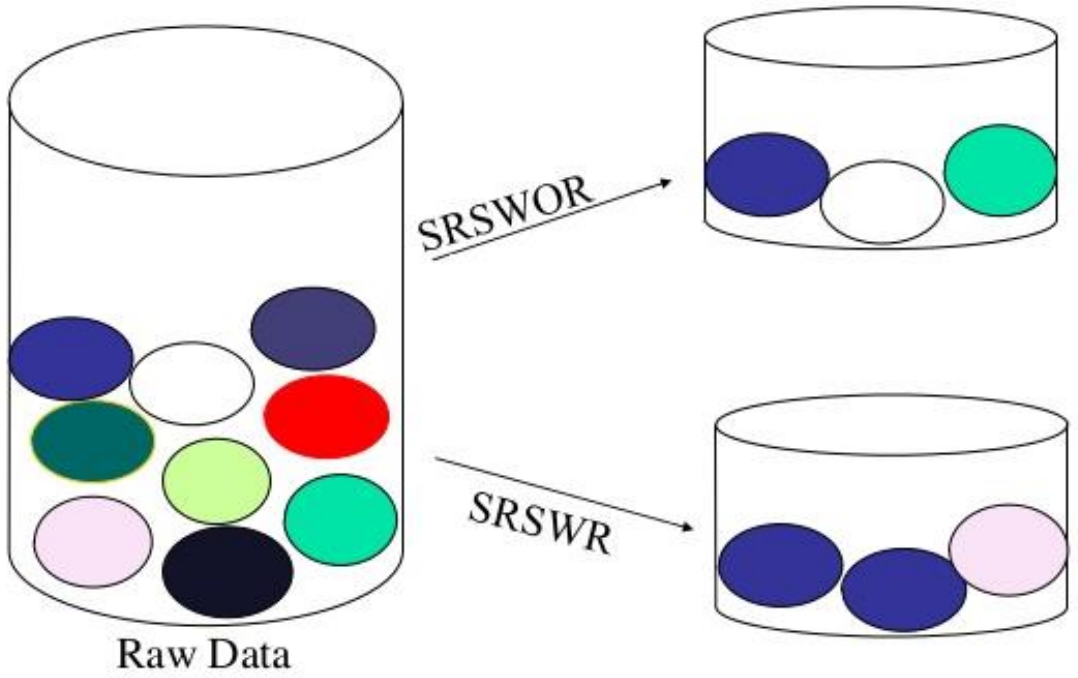
- Use a single, arbitrary learning algorithm but resample training data to make it learn multiple models parallelly.
  - ❖Data 1 ≠ Data 2 ≠ … ≠ Data m
  - ❖Learner 1 = Learner 2 = … = Learner m
- Bootstrap aggregating (Bagging)
  - ❖Repeatedly randomly sample the training data with replacement to generate multiple new training sets.



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Sampling: With or Without Replacement

- Simple random sampling without replacement (SRSWOR)
- Simple random sampling with replacement (SRSWR)



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Bootstrap

- Sampling with replacement

Data ID

Original Data	1	2	3	4	5	6	7	8	9	10
Bagging (Round 1)	7	8	10	8	2	5	10	10	5	9
Bagging (Round 2)	1	4	9	1	2	3	2	7	3	2
Bagging (Round 3)	1	8	5	10	5	5	9	6	3	7

- At each time, each example has probability *1/n* of being selected
- By sampling with replacement, some data may be repeated and some data may not appear in each bootstrap sample.

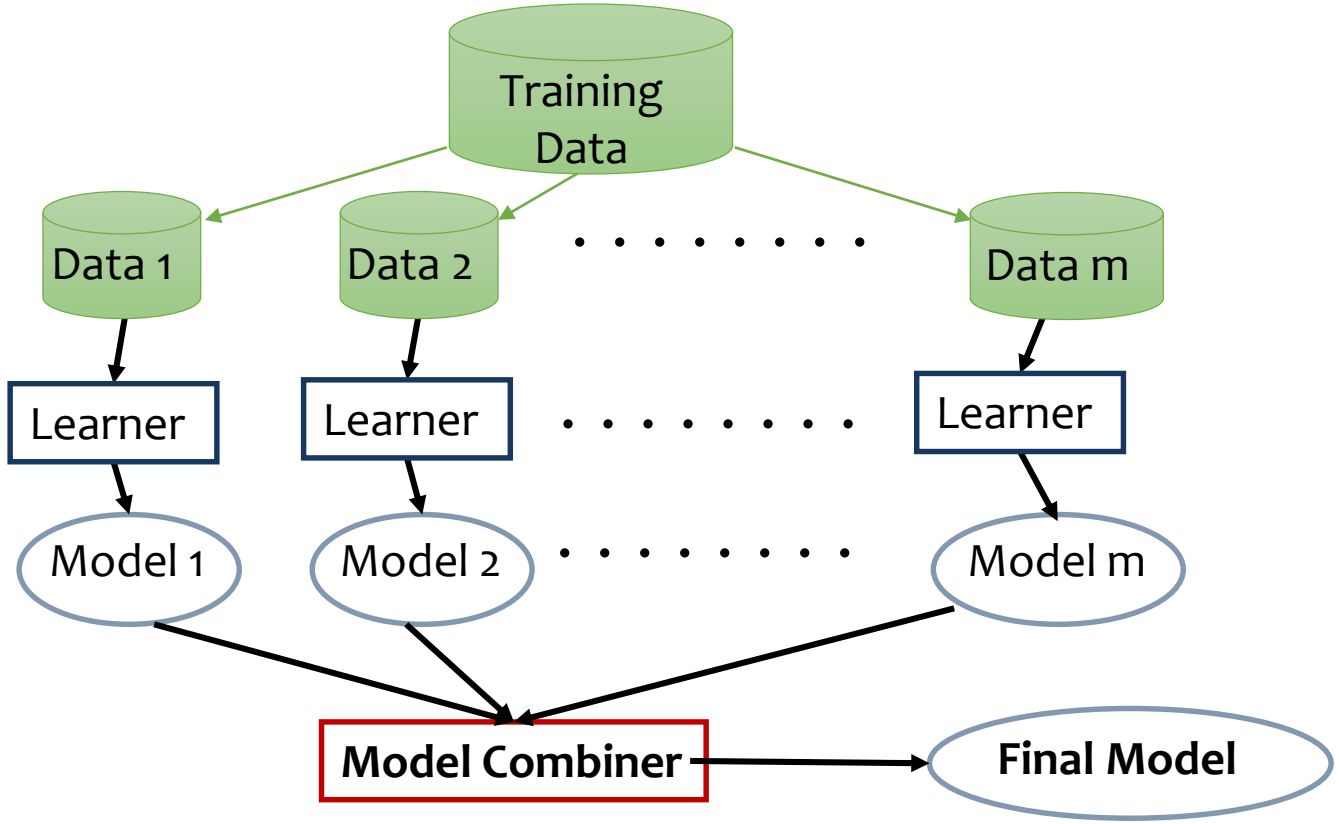


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Bagging

1) Given a training set of size *n*, create *m* bootstrap samples of size *n*

2) Using the same learner (e.g., decision tree) to train a model for each bootstrap sample and combine the *m* resulting models using simple majority vote.



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When does Bagging Work?

- Learning algorithm is unstable if small changes to the training set cause large changes in the learned classifier.
- If the learning algorithm is unstable, then Bagging almost always improves performance
  - ❖Unstable classifiers: e.g., decision trees, linear regression



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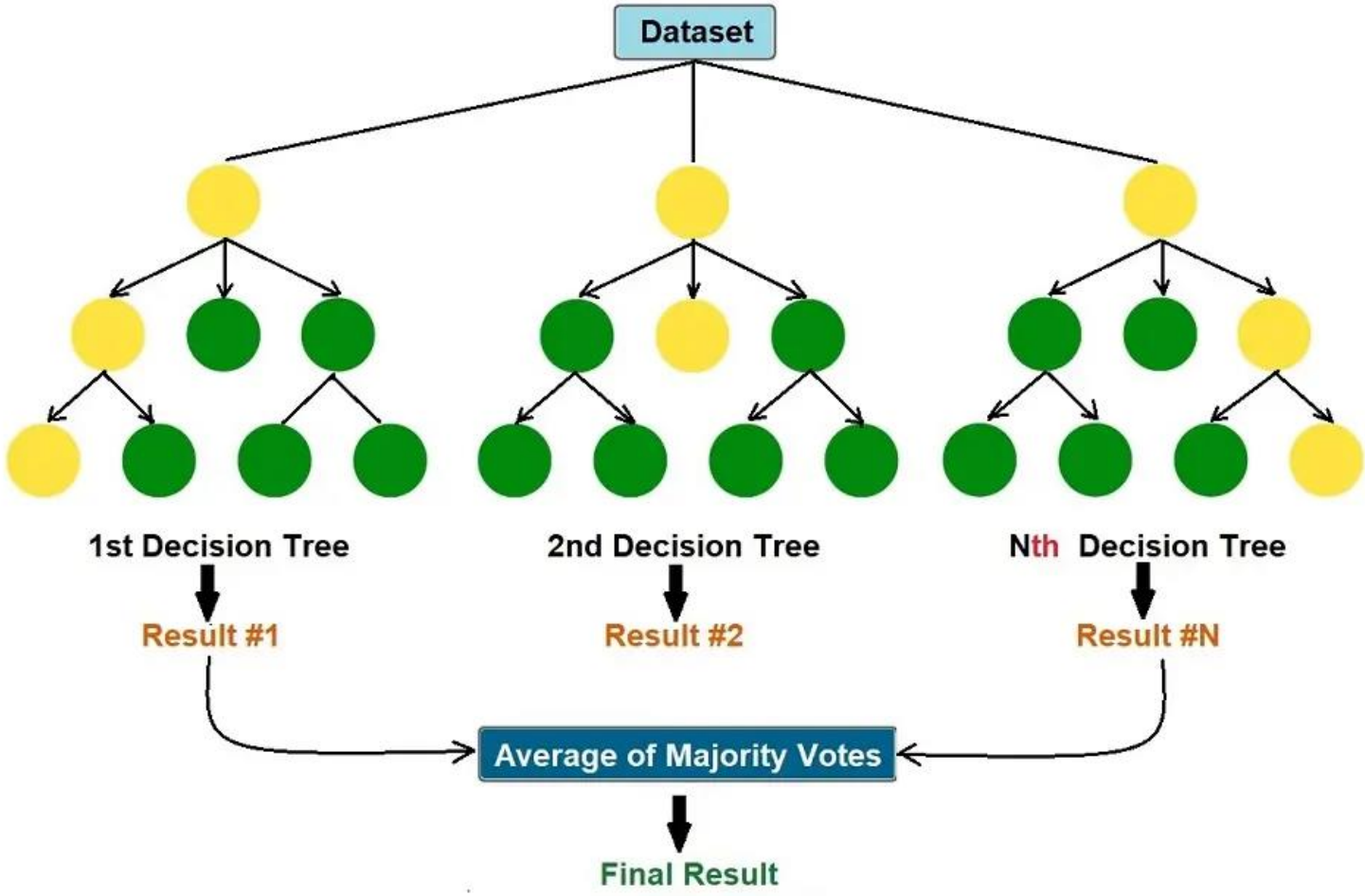
Random Forest

- Problems of bagged decision trees
  - ❖With lots of data, we usually learn fairly similar classifiers for resampled data sets, e.g., the attributes selected for the first split are usually the same across trees
  - ❖Averaging over these predictions that are not diverse enough doesn't help!
- Introduce extra variation in learner
  - ❖At each split, consider only a subset of randomly selected attributes ("feature bagging")
  - ❖Enforces diversity ("best" attributes may not be available)
  - ❖No need to prune: bagging can deal with overfitting
  - ❖Average over these learners (majority vote) to aggregate results



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Random Forest



Achieve diversity via

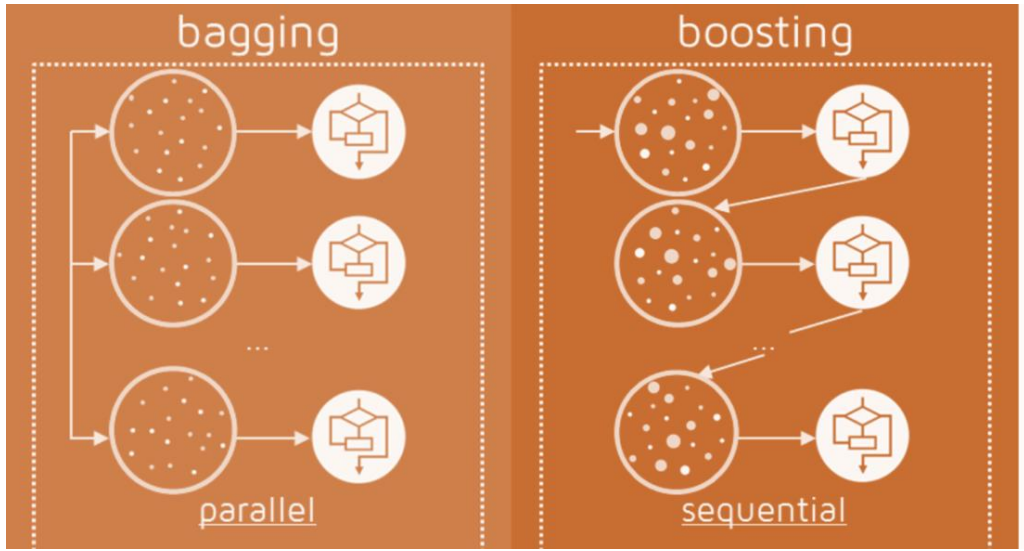
- Bootstrap training samples for each tree
- Random selection of a subset of attributes at each node



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Boosting

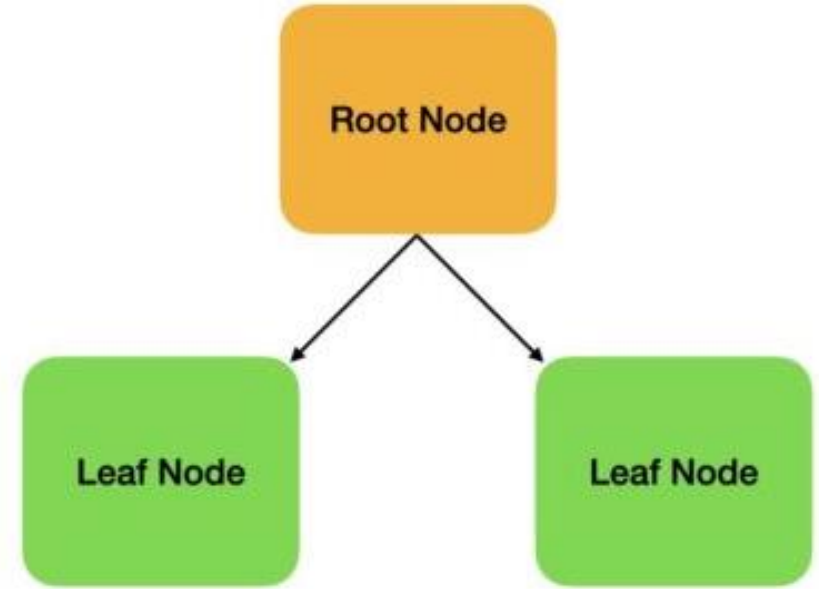
- Use a single and weak learning algorithm to learn multiple models sequentially. Each model will give more weights to examples that were mistakenly classified in the previous model.
  - ❖Learner 1 = Learner 2 = … = Learner m
  - ❖Data 1 ≠ Data 2 ≠ … ≠ Data m
- When combining the predictions from multiple models, the model with higher accuracy will have higher weights.

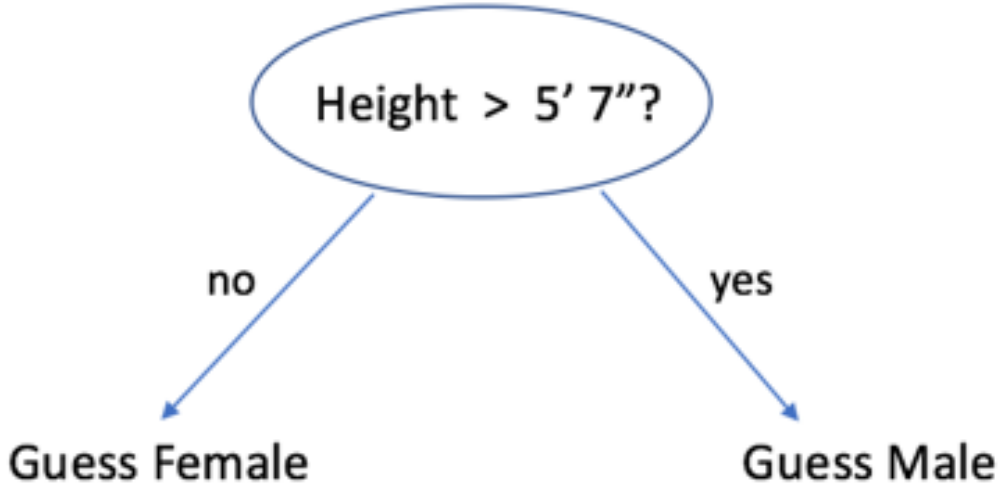


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Weak learners

- Decision stump: 1-node decision tree

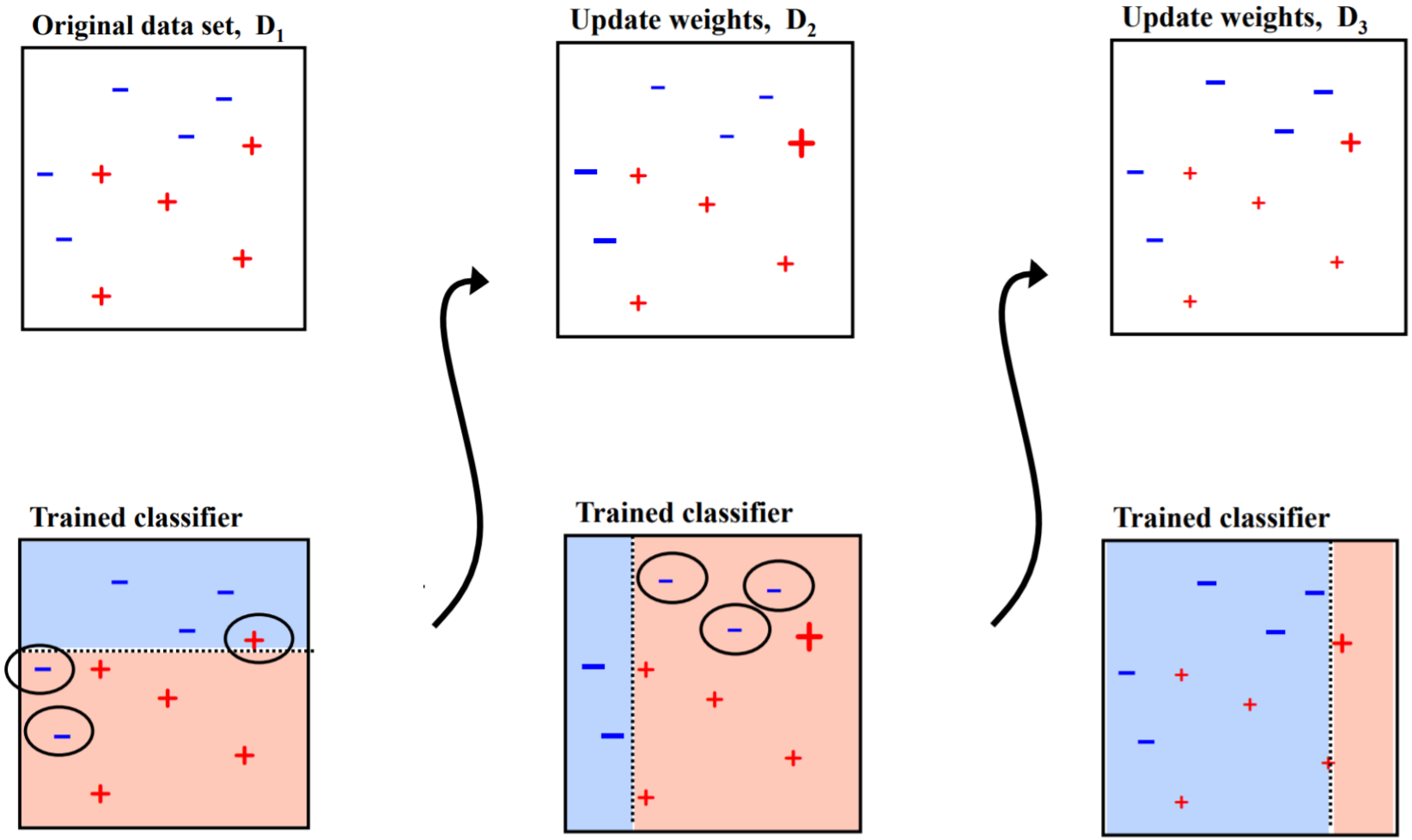




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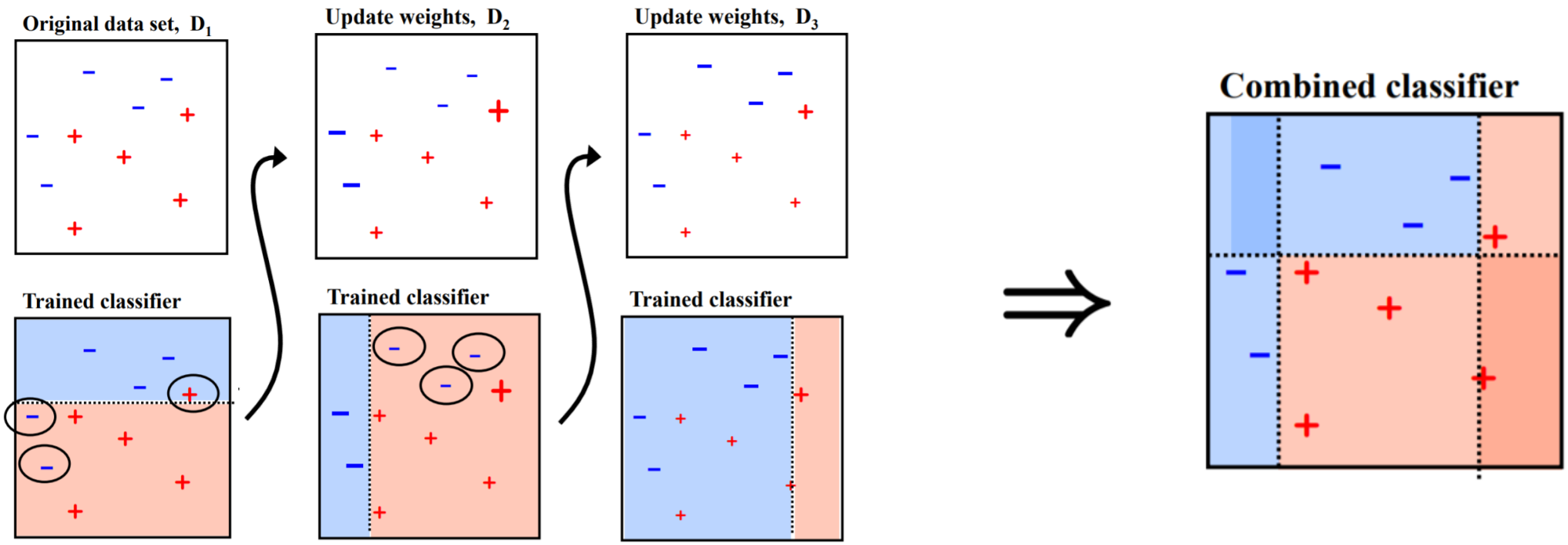
AdaBoost (Adaptive Boosting, Freund, Schapire, et al., 1996)

The idea: focus on your mistakes by training sequential models.

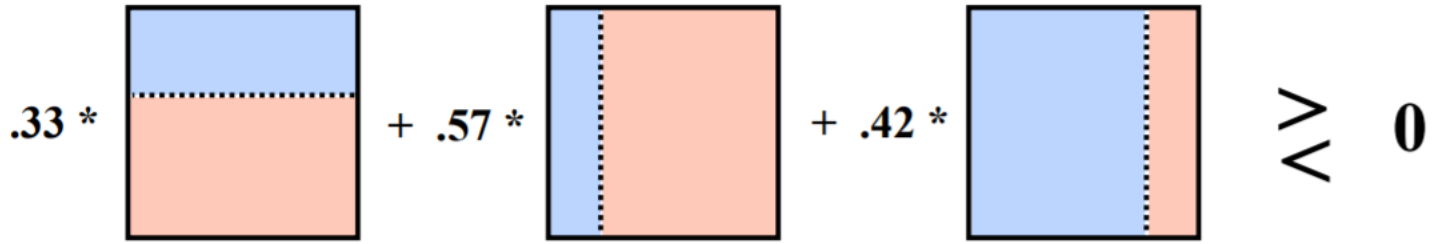


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AdaBoost-Combining models



Weight each classifier and combine them:



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AdaBoost: update weights of each data points

Initial weights

feature1	feature2	feature3	...	featureN	Y	(Weight)
data1_1	data1_2	data1_3	...	data1_N	1	(1/M)
data2_1	data2_2	data2_3	...	data2_N	0	(1/M)
...	...	...	...	...	...	...

$$err_m = \sum_{Y_i \neq T_m(x_i)} W_i$$

Total error of the classifier, sum of weights of mistake examples

$$\alpha_m = \beta \cdot \ln \frac{1 - err_m}{err_m} \quad (\beta\text{: constant})$$

Transformed total error. Smaller the total error is, the bigger α is.

$$W_i \rightarrow W_i \cdot e^{\alpha_m}$$

For mistake examples, increase weights. Increase more if the model's performance is good.

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AdaBoost: Sequential learning

- After updating the weights of data examples
  - Using method that can incorporate weights, e.g. weighted logistic regression
  - Or, just sampling data examples according to weights

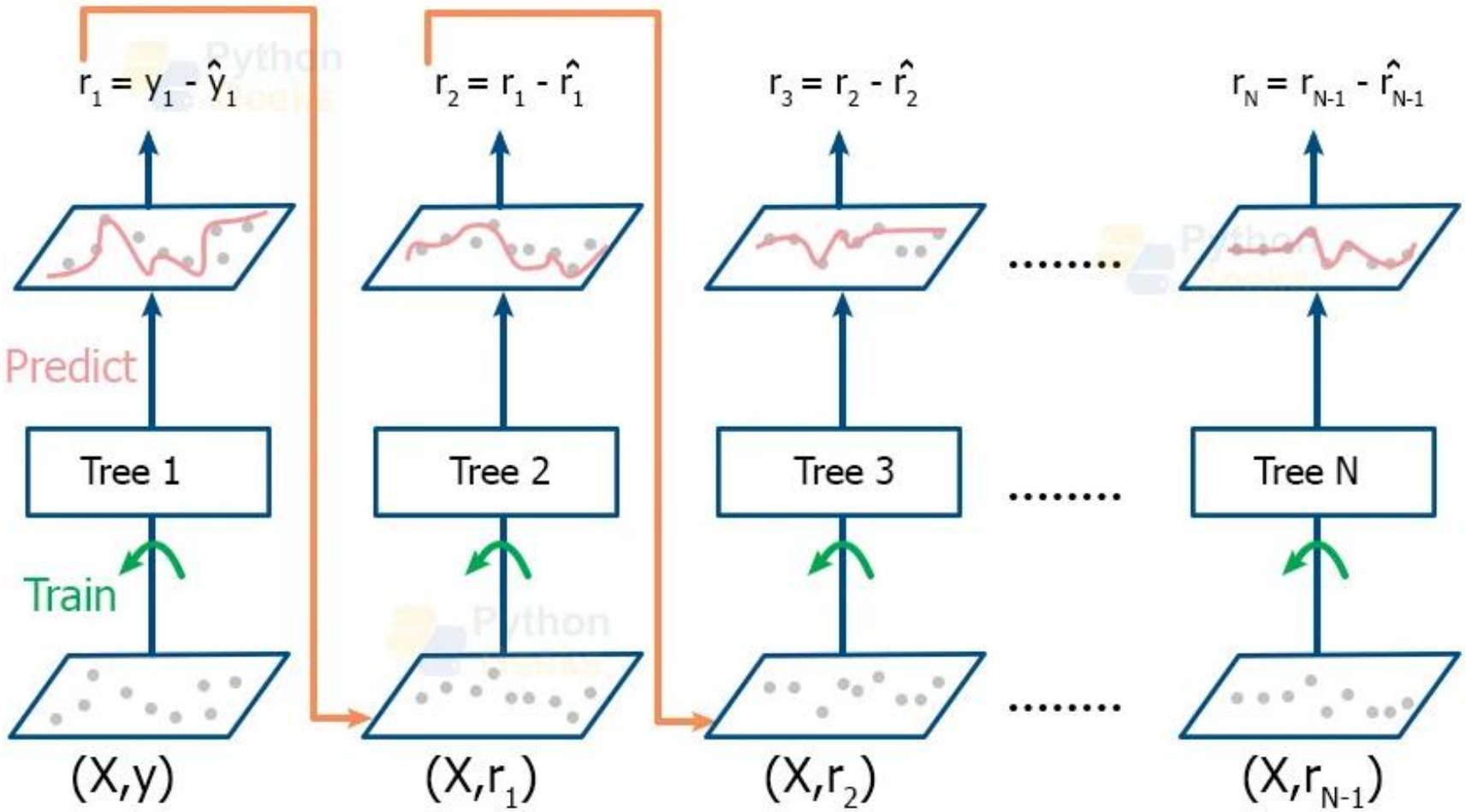
- When combining the predictions from multiple models, the model with higher accuracy will have higher weights.



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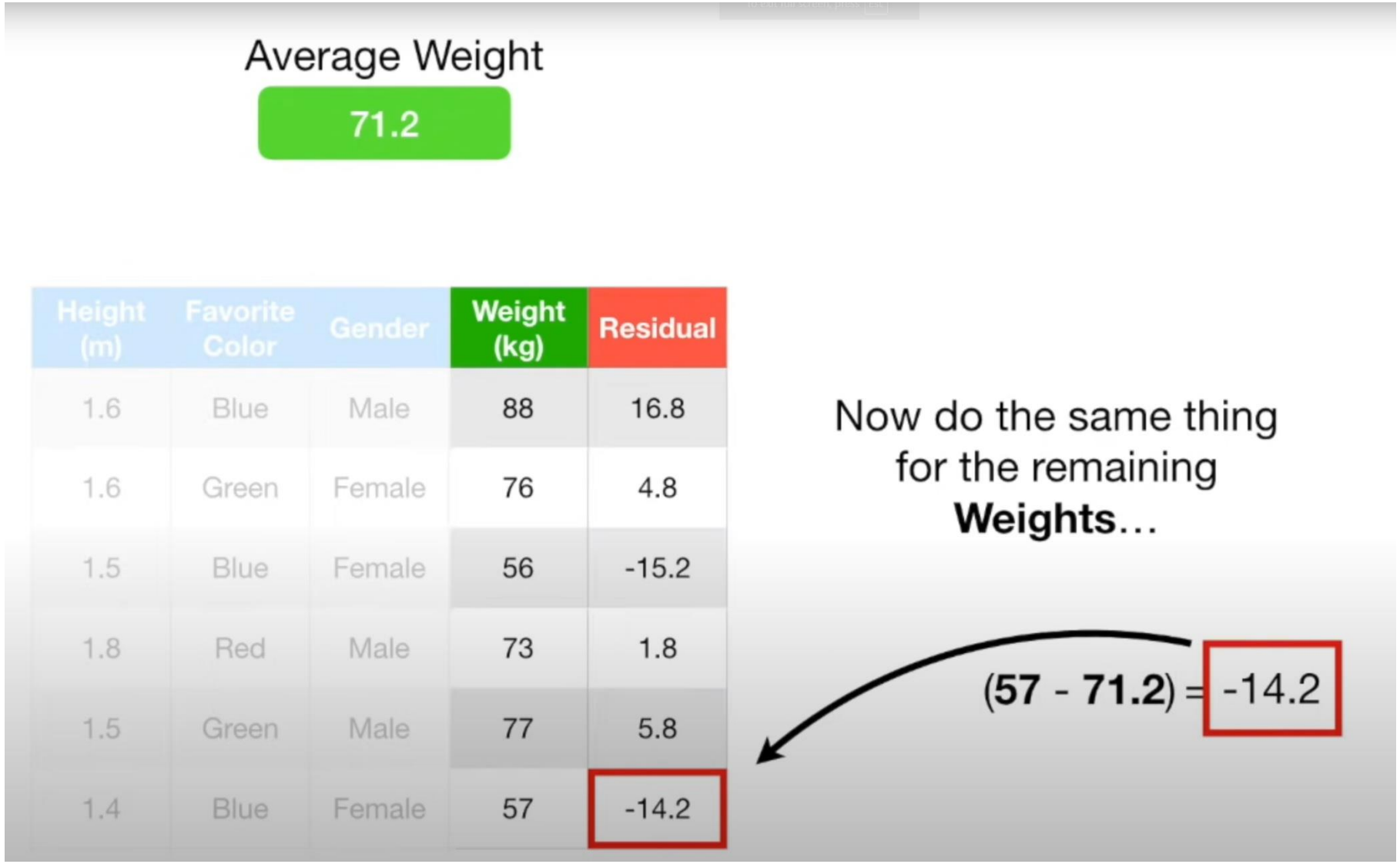
Gradient boosting (Jerome Friedman, 2001)

- Sequentially train models on the error residuals from the previous models.



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Calculating residuals



https://youtu.be/3CC4N4z3GJc?si=07eGriGjiUSffBlB

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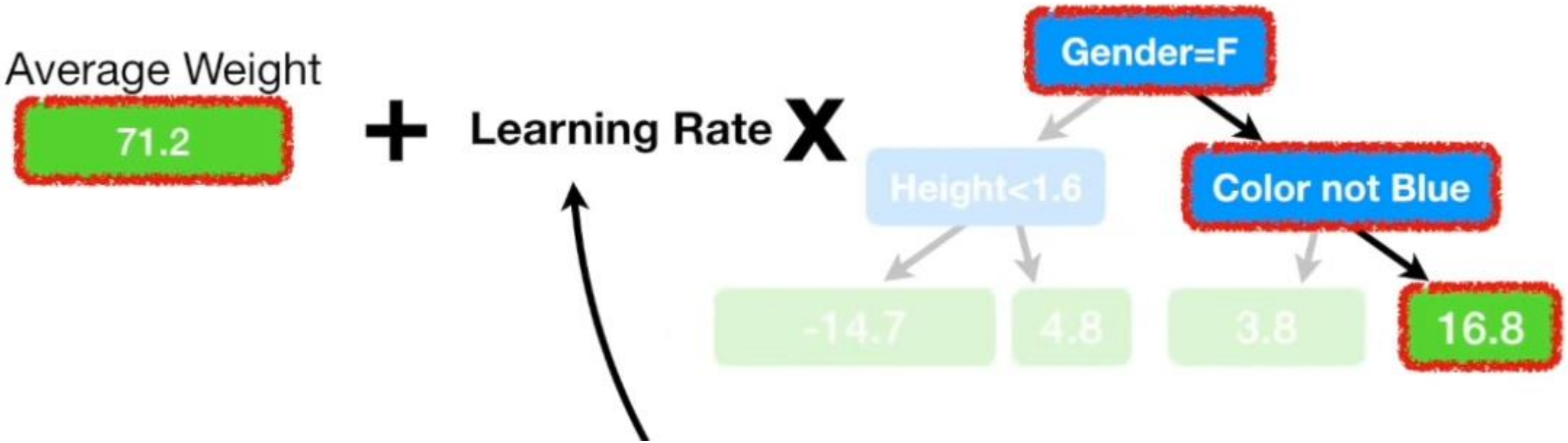
Building weak decision trees(CART) to predict residuals

Height (m)	Favorite Color	Gender		Residual
1.6	Blue	Male	88	16.8
1.6	Green	Female	76	4.8
1.5	Blue	Female	56	-15.2
1.8	Red	Male	73	1.8
1.5	Green	Male	77	5.8
1.4	Blue	Female	57	-14.2



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Update residuals

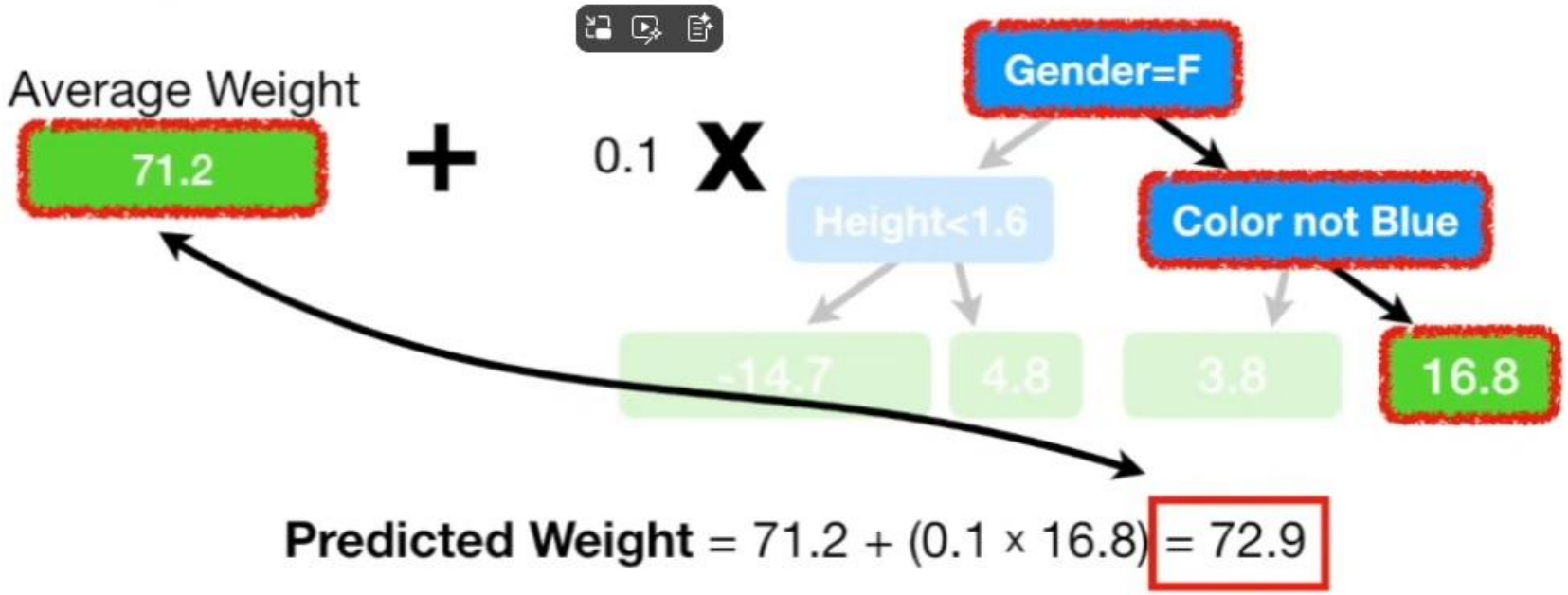


Height (m)	Favorite Color	Gender	Weight (kg)	Residual
1.6	Blue	Male	88	16.8
1.6	Green	Female	76	4.8
1.5	Blue	Female	56	-15.2
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1.5	Green	Male	77	5.8
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Height (m)	Favorite Color	Gender	Weight (kg)	Residual
1.6	Blue	Male	88	15.1
1.6	Green	Female	76	4.3
1.5	Blue	Female	56	-13.7
1.8	Red	Male	73	1.4
1.5	Green	Male	77	5.4
1.4	Blue	Female	57	-12.7

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Learning to approach the true value

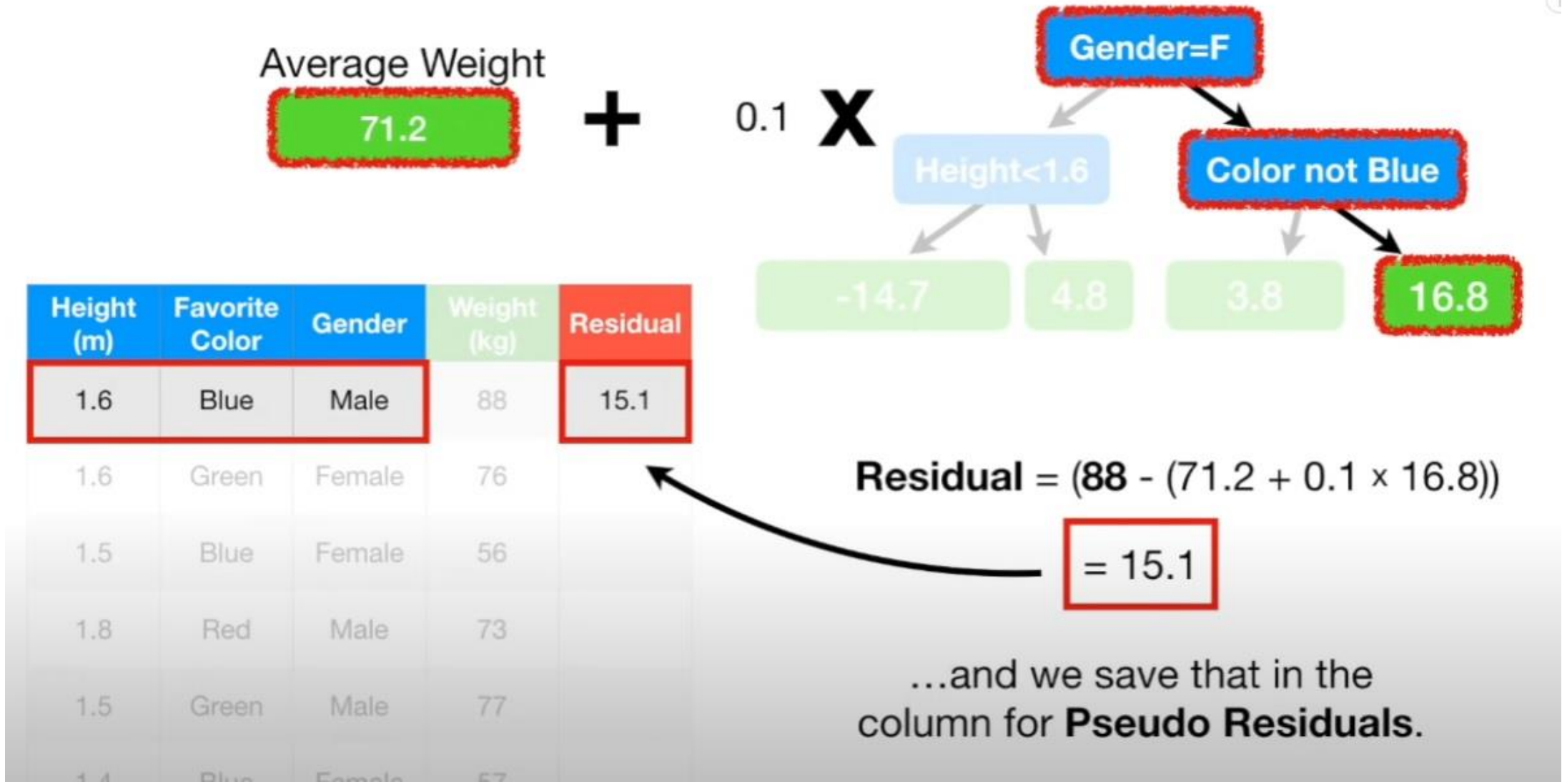


Height (m)	Favorite Color	Gender	Weight (kg)
1.6	Blue	Male	88

…but it's a little bit better than the **Prediction** made with just the original leaf, which predicted that all samples would weigh **71.2**.

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Update residuals



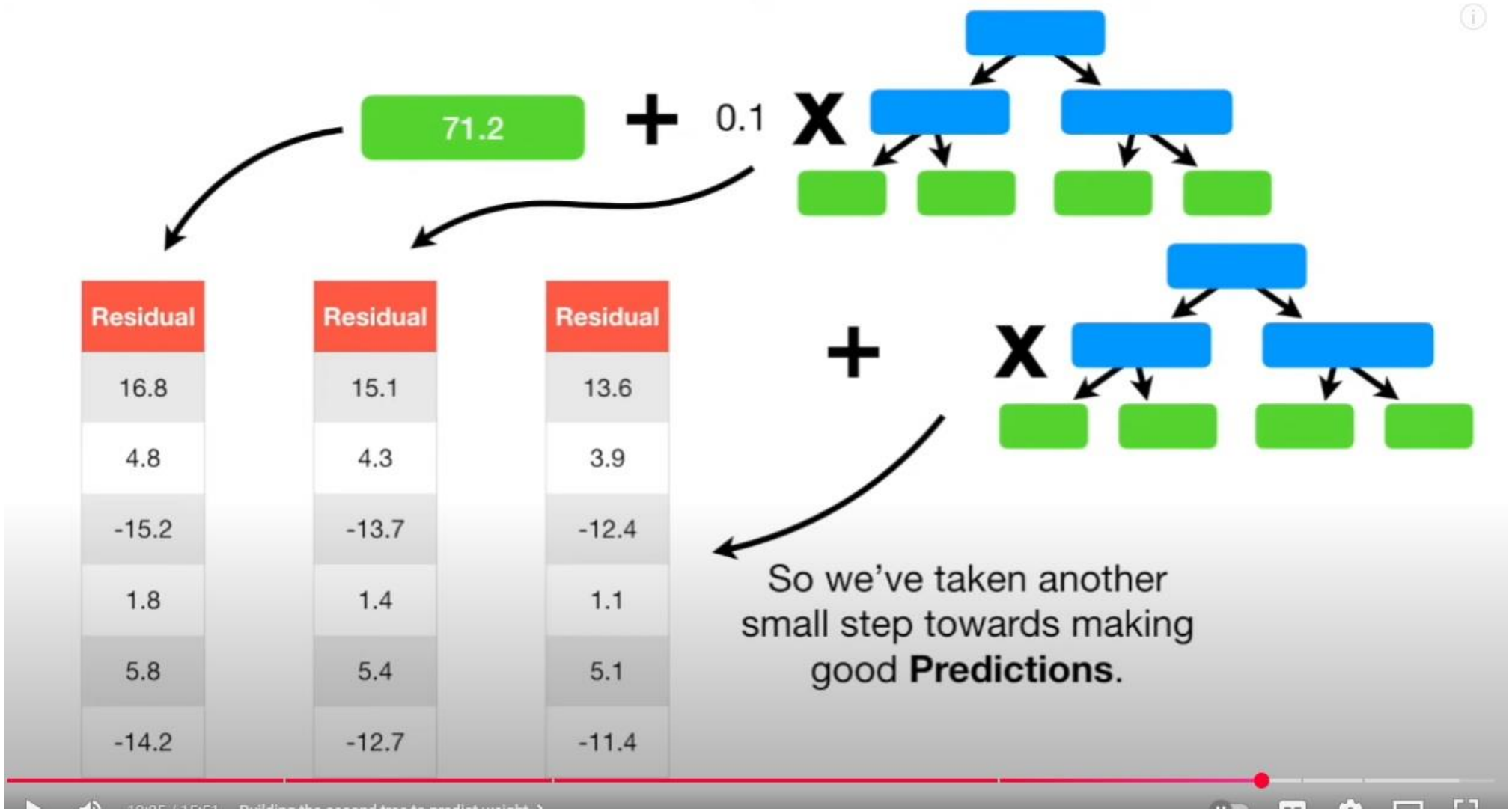
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Keep updating



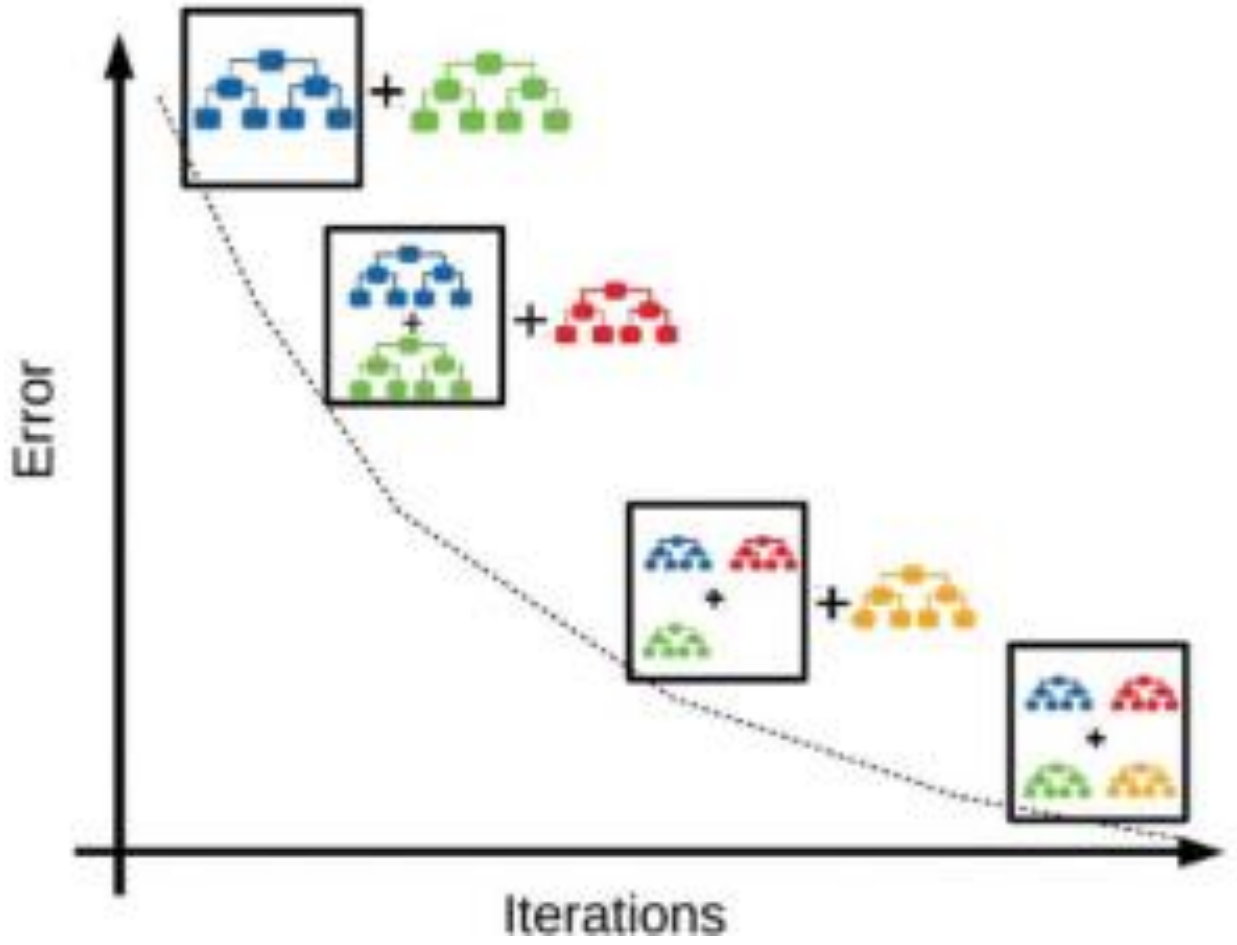
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Main idea of Gradient boosting



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Combine models



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XGBoost (e**X**treme **G**radient **B**oosting, Chen and Guestrin, 2016)

- **Faster training** - Parallel processing and optimized algorithms
- **Better generalization** - Built-in regularization prevents overfitting
- **Handles missing data** - No need for imputation
- **More efficient** - Better memory usage and computational optimization
- **Advanced split finding** - Weighted quantile sketch for better performance

- **Gradient Boosting**: Good baseline, simpler projects, when interpretability is key
- **XGBoost**: Large datasets, computational efficiency needed, competition settings, production systems
- **Bottom line**: XGBoost is essentially Gradient Boosting with engineering optimizations and regularization, making it faster and more robust for most real-world applications.



https://www.youtube.com/watch?v=OtD8wVaFm6E

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Pros and Cons of Ensemble Learning

- Pros
  - ❖Better prediction performance
  - ❖A major strategy for participants in data mining contests and data analysts/scientists in various companies.
- Cons
  - ❖Requires more effort and computationally expensive
  - ❖Difficult to explain (blackbox method)



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